Ko Lih Han

Software Engineer | Applied AI & Backend

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EXPERIENCE

Kahap Inc | Software Developer (Part-time) Aug 2022 - Aug 2026

- Led end-to-end development of a production feedback platform, from system design and implementation through deployment, including submission workflows, automated notifications, and admin operations.
- Shipped new features and resolved application and database issues on an existing Laravel/Vue.js production platform.
- Managed server migrations and upgrades, Laravel queues/workers/schedulers, SSL renewal, and production database troubleshooting.

Pac Com Solution | Full-Stack Developer Intern Jun 2021 - Jan 2022

- Developed and maintained Laravel/Vue.js features for a production POS system.

PROJECTS

Text-to-SQL with Verification & Repair

Python | FastAPI | LangGraph | SQLite | Qwen3.5 9B

- Built a local Text-to-SQL system that uses database schema and values to generate SQL, validates queries before execution, and repairs them when needed.
- On 100 BIRD DEV queries, verification and repair increased executable queries from 75 to 90 and recovered 6 additional correct queries.

Hybrid Retrieval for RAG

Python | FastAPI | BM25 | Qwen3-Embedding-0.6B | RRF

- Built a hybrid retrieval service for RAG that combines keyword and semantic search, using RRF to merge the results.
- On 64 EnterpriseRAG-Bench queries over 5,189 documents, hybrid retrieval improved Recall@10 from 0.734 to 0.763 and MRR@20 from 0.679 to 0.742 versus BM25 alone.

Multimodal Video QA Agent

Python | Streamlit | Qwen3.5 4B | FFmpeg

- Built a local video QA agent that routes each question to transcript retrieval or frame/clip inspection and shows the supporting evidence.
- In a 12-question evaluation, the agent correctly answered 4 of 6 questions requiring visual evidence and avoided unnecessary visual-tool calls on transcript-answerable questions.

M.S. Thesis: Interpreting Implicit Intent through Workplace Hierarchies and Context

- Built a graph-aware retrieval system over 10,645 workplace dialogue turns, using workplace roles and conversation context to find evidence for interpreting implied intent.
- On 400 held-out examples, role-based retrieval improved semantic F1 from 0.6440 to 0.6793; combining role and context evidence reached 0.6819.

SKILLS

AI / Retrieval: RAG, Information Retrieval, LLM Evaluation, Agents, Tool Calling, LangGraph
Programming / Backend: Python, SQL, JavaScript, PHP, FastAPI, REST APIs, Laravel, Vue.js
Databases / Dev Tools: MySQL, SQLite, Linux, Git, GitHub Actions, pytest
Languages: Chinese (Native), English (Fluent), Malay (Intermediate)

EDUCATION

National Tsing Hua University | M.S., Information Systems & Applications Sep 2024 - Jun 2026

GPA 4.09/4.3 | Taiwan

Chung Hua University | B.S., Computer Science & Information Engineering Sep 2022 - Jun 2024

GPA 3.9/4.0 | Taiwan